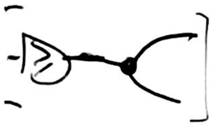
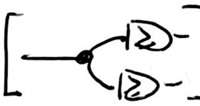
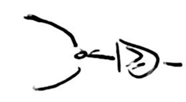
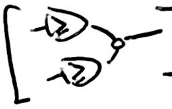
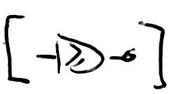
$\left[\right.$  $\left.\right] = \left\{ \left(x, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right) \middle| \; x \geq y_1 \wedge x \geq y_2 \wedge y_1 = y_2 \right\}$

$\subseteq \left\{ x, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \middle| \; x \geq y_1 \wedge x \geq y_2 \right\} =$ $\left[\right.$  $\left.\right]$

$\left[\right.$  $\left.\right] = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, y \middle| \; x_1 + x_2 \geq y \right\} =$

$\left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, y \middle| \; \overset{\exists c, d}{x_1 + x_2 \geq c + d} \wedge c + d = y \right\} =$

$\left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, y \middle| \; \exists \hat{c}, \hat{d} \;\; x_1 \geq \hat{c} \wedge x_2 \geq \hat{d} \wedge \hat{c} + \hat{d} = y \right\} =$ $\left[\right.$  $\left.\right]$

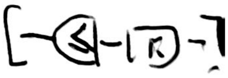
$\left[\right.$  $\left.\right] = \{ (x, \bullet) \mid \exists c \;\; x \geq c \} \overset{\text{Tautologia}}{=} \{ (x, \bullet) \mid x \in \mathbb{R} \} =$ $\left[\right.$  $\left.\right]$

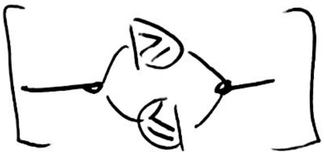
$\left[\right.$  $\left.\right] = \{ (x, \bullet) \mid x = 0 \}$ $\{ (x, \bullet) \mid x \geq 0 \wedge x = 0 \}$

$\subseteq \{ (x, \bullet) \mid x \geq 0 \} =$ $\left[\right.$  $\left.\right]$

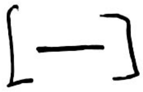
$>0)$ $\left[\right.$  $\left.\right] = \{ (x, y) \mid kx \geq y \wedge k > 0 \} = \{ (x, y) \mid x \geq \frac{y}{k} \wedge k > 0 \}$

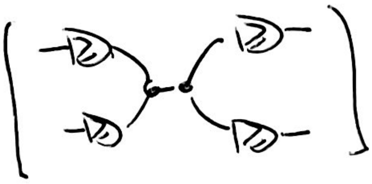
$= \{ (x, y) \mid \exists c \;\; x \geq c \wedge kc = y \wedge k > 0 \} =$ $\left[\right.$  $\left.\right]$

$k < 0)$ $\left[\right.$  $\left.\right] =$ Same as above, flip the $\geq$ $=$ $\left[\right.$  $\left.\right]$

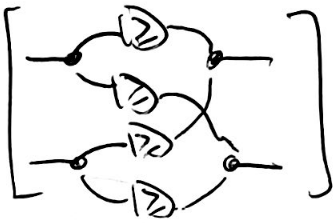
 $= \{(x,y) \mid x \geq y \wedge x \leq y\} =$

$\{(x,y) \mid x = y \wedge x,y \in \mathbb{R}\} \subseteq \{(x,y) \mid x = y \wedge x,y \in \mathbb{R}\}$

$=$ 

 $= \{((x_1,x_2),(y_1,y_2)) \mid \exists z \;\; x_1 \geq z \wedge x_2 \geq z \wedge z \geq y_1 \wedge z \geq y_2\}$

$= \{((x_1,x_2),(y_1,y_2)) \mid x_1 \geq y_1 \wedge x_1 \geq y_2 \wedge x_2 \geq y_1 \wedge x_2 \geq y_2\}$

$=$ 

 $= \{(x,y) \mid \exists z \;\; z \geq y \wedge z \geq x\} =$

$= \{(x,y) \mid x,y \in \mathbb{R}\} =$